Surface Compositional Analysis of Venus via VIRTIS-M-IR Data and Multi-Layered Radiative Transfer Modeling

Objective

.This research addresses a fundamental gap in planetary science: the lack of high-resolution, global compositional data for the Venusian surface. Due to the optically thick CO2 atmosphere and dense sulfuric acid cloud decks, the Venusian surface remains largely opaque to remote sensing. This project develops a python computational pipeline to retrieve surface emissivity by inverting hyperspectral nightside thermal emission data acquired by the VIRTIS-M-IR instrument onboard the Venus Express mission. By isolating the subtle surface-emitted radiance escaping through "spectral windows," we aim to characterize the mineralogical composition and geologic evolution of the Venusian crust.

Methodology

The study utilizes hyperspectral datasets acquired by the VIRTIS-M-IR instrument onboard the Venus Express mission with certain constraints such as nocturnal data, low angle of observation and >3sec time exposure to obtain the best possible datasets for our pipeline following Muller et al.(2020). Given the high opacity of the Venusian atmosphere, the focus remains exclusively on nocturnal observation prospects, specifically targeting the three “surface windows” of 1.02, 1.10 & 1.18 mu m and two “atmospheric windows” of 1.74 & 2.30 mu m where cloud opacity is sufficiently low to allow thermal photons from the surface to escape.

Our methodology centers on the development of a fully coupled, multi-stream radiative transfer (RT) forward model. We invert the VIRTIS radiance cubes by simulating the light path of thermal photons originating at the crust-atmosphere interface, undergoing multiple scattering within the cloud decks, and eventually being detected at the Top of Atmosphere (TOA). The retrieval pipeline centers on a high-fidelity Radiative Transfer (RT) solver based on the formalism established by Korkin et al.(2022). The forward model incorporates the following parameters:

- Atmospheric Structure: We utilize the Venus International Reference Atmosphere (VIRA) to constrain local vertical pressure-temperature profiles, constrained further by Magellan-derived altimetry data. This allows for precise calculation of the hydrostatic gas distribution and surface temperature estimation at pixel-specific coordinates.
- Aerosol Scattering: We adopt the 4-mode cloud structure following Häus et al. (2014), which improves our model by incorporating a transitionary "haze" layer (Mode 2') and an optimized Mode 3 basal layer. We employ the Henyey-Greenstein phase function to characterize the strongly forward-scattering aerosol particles, essential for accurate radiance propagation in the 1.0 to 2.5 mu m range.
- Line-by-Line Spectroscopy: Utilization of the HITRAN/HITEMP/HOTBASE database to compute accurate gas opacity matrices, applying non-LTE corrections for upper-atmospheric state populations where relevant, and sub-Lorentzian chi-factor corrections for CO2 far-wing absorption at deep-crustal pressures up to 90 bar.

Data Processing & Retrieval

The computational architecture is designed for massive parallelization. We utilize a line-by-line gas matrix compilation method optimized via HAPI, distributed across high-performance CPU clusters to minimize processing time for the full VIRTIS hyperspectral grid.
The core of our retrieval pipeline is an iterative solver utilizing a discrete-stream Gauss-Seidel method. This engine performs a closed-loop coupling of upwelling radiance and downwelling diffuse "sky-glow." By calculating the downwelling flux at the surface boundary, we correctly account for the component of the radiation field reflected by the surface (based on Fresnel and Lambertian approximations). Our retrieval algorithm minimizes the residual radiance (observed - simulated) by iterating through a parameter of cloud opacity to match the simulated radiance with that of the observed one at the atmospheric windows where we assume nil surface emission, thenkeeping the obtained cloud opacity we iterate for surface emissivity at the surface windows.
The pipeline employs a closed-loop coupling of upwelling and downwelling diffuse radiation fields to account for the reflection of atmospheric "sky-glow" from the surface. Synthetic spectra are generated for each target pixel by simulating the complete column-integrated radiance, which is then subtracted from the observed VIRTIS radiance. The residual pattern provides the direct signature of surface emissivity.

Expected Outcomes

The primary output is a map of surface emissivity at the $1.02, 1.10, 1.18, 1.74,$ and $2.30\ \mu m$ windows. By correlating these emissivity signatures with laboratory-measured spectra of igneous analogs (such as basalt, andesite, and rhyolite) from Crank et al.(2007), this research will produce definitive compositional constraints for the target regions.

This research will provide critical insights into:

- Magmatic History: Mapping the spatial distribution of basaltic vs. evolved felsic crusts to test the hypothesis of localized crustal recycling or silica-rich continents ("tesserae").
- Chemical Weathering: Identifying surface-atmosphere interactions by assessing the degree of alteration in high-altitude volcanic shields.
- Future Missions: This validated pipeline serves as a modular framework for future high-resolution missions, including NASA's DAVINCI+ and ESA's EnVision.

By integrating state-of-the-art spectroscopic corrections, the Häus cloud model, and high-performance RT numerical solvers, this work moves beyond traditional observational analysis to provide a quantitative, physically-based mineralogical interpretation of the Venusian surface.

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